Metrological Domain Profiling II:
Functional Economic Ontologies from the Aegean Bronze Age to the Inca Empire

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Abstract

Metrological Domain Profiling (MDP), a computational method for classifying signs in undeciphered administrative scripts by their co-occurrence with commodity ideograms and numerals, is extended from its original three-script validation (Proto-Cuneiform, Proto-Elamite, Indus Valley) to four additional writing systems spanning two new continents and an entirely new recording medium. Applied to the Aegean script family — Cretan Hieroglyphic (~2100 BCE), Linear A (~1800 BCE), and Linear B (~1400 BCE) — MDP traces the 700-year evolution of Minoan-Mycenaean administrative vocabulary and reveals that metrological domain separation was already established in the earliest Cretan writing. Applied to the Inca Khipu (~1400 CE), MDP demonstrates that the same principle operates on knotted textile cords, demonstrating applicability across media (clay, stone, string), continents (Asia, Europe, South America), and 4,600 years of administrative history. When applied to deciphered Linear B as a control, MDP achieves low semantic agreement (14%) with Greek dictionaries but high administrative-context accuracy — correctly identifying that Mycenaean scribes administered slaves through the grain department. This "functional ontology" finding suggests MDP reveals administrative reality rather than linguistic meaning, providing economic historians with a tool that pure decipherment cannot replicate.

Keywords: *computational epigraphy, undeciphered scripts, Linear A, Linear B, Cretan Hieroglyphic, Inca Khipu, metrological systems, administrative archaeology, Bronze Age economy*

1. Introduction

1.1 Background and Motivation

Administrative record-keeping is one of the oldest functions of writing. From the proto-literate tablets of 4th-millennium Mesopotamia to the knotted cords of the pre-Columbian Andes, human societies that reached a certain threshold of organisational complexity independently developed systems for tracking commodities, labour, and resources. These systems share a fundamental structural property: they pair signs denoting *what is being counted* with signs denoting *how many*. This pairing creates a diagnostic signature that can be exploited computationally, even when the commodity signs remain undeciphered.

In Sharman (2026), we introduced Metrological Domain Profiling (MDP), validated against Proto-Cuneiform (100% accuracy on 28 known signs), applied to Proto-Elamite (79 signs classified), and extended to the Indus Valley Script (where it identified "semantic locks"). The present paper extends MDP to four additional scripts, completing a seven-script portfolio spanning four continents, three recording media, and 4,600 years.

1.2 The Golden Rule of MDP

MDP operates under a strict constraint: **it only works on administrative ledgers**. The method requires numerals and commodity identifiers on the same document. It will not work on religious texts, royal monuments, or narrative inscriptions. This ensures results only where data supports them.

1.3 Scope and Contributions

1. **The Aegean Trilogy**: Cretan Hieroglyphic, Linear A, and Linear B — tracing 700 years of administrative vocabulary evolution.
2. **The Linear B Control**: MDP classifies by *administrative context* not *semantic meaning* — the "functional ontology" discovery.
3. **The Khipu Extension**: 619 khipus, 54,403 cords, 110,677 knots — proving the principle operates on knotted textile cords.
4. **The Administrative Segregation Principle (ASP)**: A strong recurring pattern across seven scripts, four continents, three media, 4,600 years.
5. **Measure Subunit Divergence**: T/V for dry grain, S for liquid oil, M/N/P for weight — confirmed by Linear B decipherment.

2. Data Sources and Corpora

Script	Period	Corpus Size	Source	Status
Cretan Hieroglyphic	2100-1700 BCE	317 inscriptions 1,198 tokens 434 unique signs	linear0.xyz (Hogan)	Undeciphered
Linear A	1800-1450 BCE	1,712 inscriptions 419 ledger tablets 763 unique words	lineara.xyz (Hogan)	Undeciphered
Linear B	1400-1200 BCE	5,832 inscriptions 62,148 tokens 5,823 unique words 65 commodity ideograms	linearb.xyz (Hogan)	Deciphered (control)
Inca Khipu	1400-1532 CE	619 khipus 54,403 cords 110,677 knots 8,038,367 total value	Open Khipu Repository	Partially understood

Table 1. Corpora analysed in this study. All sources are open-access.

Linear B sites: Knossos (4,223), Pylos (986), Thebes (363), Mycenae (87). Khipu provenance: Pachacamac (90 khipus, 4,612 cords), Incahuasi (52, 3,703 cords), Leymebamba (22, 3,214 cords), Ica (49, 2,301 cords). Dominant khipu fibre: cotton (28,895 cords).

3. Methodology

The core MDP algorithm (Sharman, 2026) operates in three steps: (1) **Token Classification** (WORD, COMMODITY, NUMERAL, MEASURE, STRUCTURAL); (2) **Co-occurrence Profiling** across all inscriptions; (3) **Domain Classification** (>60% = dominant, 40–60% = LEANING, <35% across multiple = CROSS-COMMODITY).

Aegean scripts: Standard MDP with register separation for Linear A. **Linear B**: Dual-metric validation (semantic + administrative-context). **Khipu**: Cord colour = commodity, knot value = numeral, knot type = measure indicator, ply direction = domain marker.

4. Results

4.1 Cretan Hieroglyphic: The Grandmother Script

317 inscriptions (Knossos: 68, Malia: 63). Only 17 contain commodity ideograms and 46 contain numerals, reflecting the corpus's predominance of short seal impressions.

KU-RO cognate confirmed. The sign KU-RO (“total” in Linear A) appears twice with 50% numeral adjacency — the earliest known administrative summation function in Aegean writing, inherited by Linear A across three centuries.

Sign	Freq	Num. Adj.	Co-occurrence	Interpretation
*161	4	100%	none	Probable commodity ideogram
*157	2	100%	none	Probable commodity ideogram
*156	6	67%	none	Likely commodity ideogram
*155	7	57%	none	Likely commodity ideogram
*153	5	20%	GRAIN=1	Potential grain marker
*152	6	33%	LIVESTOCK=1	Potential livestock marker

Table 2. Undeciphered numbered signs identified by numeral adjacency.

Dominant domain: LIVESTOCK (14 signs). KO-RO3 (64 occurrences, 59 tablets) is a livestock-associated name or title. 30 syllabic signs shared with Linear A, confirming script family continuity.

4.2 Linear A: The Minoan Administrative Lexicon

1,712 inscriptions register-separated: Nodules (898), Ledger tablets (419, used for MDP), Roundels (151), Ritual vessels (223). 763 unique tablet words; 120 classified into commodity domains.

Register separation proved critical: KU, KA, SI, RO are 89–98% nodule stamps. After filtering, NI emerges as the top tablet word (68 occ., 0% nodules), a CROSS-COMMODITY generalist (GRA=19, OLE=16, VIN=18, CYP=21).

Domain	Words	Key Examples
CROSS-COMMODITY	33	NI (68), SA-RA2 (20), KI-RO (16, "deficit")
GRAIN	13	DA-ME (4), KU-NI-SU (5), MI-NU-TE (4), A-KA-RU (3, heading)
PERSON/MAN	11	DA-RE (6, "personnel"), TI (8), TA-NA-TI (4)
WINE	9	DI-NA-U (5), SA-RO (4), PA-DE (3)
OLIVE OIL	3	JE-DI (4), QA-*310-I (4)
NO_COMMODITY	14	U (4), SA-MA (3), *21F-TU-NE (3)
LEANING/WEAK	34	Partial signals across multiple domains

Table 3. Linear A domain classifications (120 words across 419 ledger tablets).

Two-tier summation: KU-RO (36 occ., 94% num. adj.) at MIDDLE position (subtotals); PO-TO-KU-RO at LAST_LINE (grand totals). 5 exact matches, 11 close (±2) out of 30 checked. **A-KA-RU**: 100% FIRST_LINE — a document heading. **Fraction divergence**: Grain = halves/quarters; Oil = halves/thirds/fifths; Cypress = thirds/quarters/eighths.

Scribe/room specialisation: KH Scribe 2 = 83% CYPRESS. Portico 11 = 100% OLE. This computationally derived specialisation aligns with the physical archaeology of Aegean palaces, which featured dedicated oil magazines and commodity-specific storage — MDP recovers the administrative architecture from the mathematics alone. **Libation formula** A-TA-I-*301-WA-JA: auto-classified NO_COMMODITY (0% signal, 91% FIRST_LINE on ritual vessels) — MDP cleanly separates religious from administrative content.

4.3 Linear B: The Functional Ontology Discovery

5,832 inscriptions (Knossos: 4,223; Pylos: 986; Thebes: 363). 62,148 tokens, 5,823 unique words, 65 commodity ideograms. Top commodities: OVIS:m (1,070), VIR (864), GRA (584), LANA (415). 200 words classified into 29 domain categories.

Word (occ.)	Greek	Meaning	MDP Domain	Why
pa-ro (272)	paro	from/at	GRAIN	Land-tenure formula (E-series)
ko-wo (218)	korwos	boy	LEAN_WOMAN	Children on workforce tablets
pe-mo (201)	—	—	GRAIN	Seed-corn allocations
o-na-to (187)	—	—	GRAIN	Land-lease (100% grain co-occ.)
do-e-ro (90)	doelos	slave	GRAIN	Slaves = grain ration records
ka-ke-we (52)	—	—	BRONZE	Bronze-workers (18/20 AES)
ka-ko (18)	khalkos	bronze	BRONZE	Pure commodity term ✔
ta-ra-si-ja (24)	—	—	BRONZE	Bronze allocation (16/21 AES)

Table 4. Selected Linear B classifications. MDP maps administrative context, not dictionary meaning.

Ground truth validation: Land Tenure (59 tabs) → GRAIN dominant (269 hits). Livestock (17 tabs) → LIVESTOCK (29). Industrial Production (36 tabs) → BRONZE (20). Religion/Ritual (27 tabs) → OIL (12). Military Equipment (38 tabs) → NO_COMMODITY (21). Vessels/Furniture (19 tabs) → NO_COMMODITY (18).

Domain	Primary	Secondary	Tokens	Interpretation
GRAIN	T (64%)	V (34%)	724	Dry volume
OIL	S (48%)	V (44%)	151	Liquid volume
WINE	V (50%)	S (30%)	76	Liquid (mixed)
BRONZE	M (71%)	N (28%)	276	Weight
GOLD	P (59%)	N (37%)	27	Fine weight
WOOL	M (79%)	—	63	Weight
FLOUR	V (68%)	Z (15%)	74	Fine dry volume

Table 5. Measure subunit divergence confirmed by decipherment. T=dry, S=liquid, M/N/P=weight.

Scribe	Records	Primary Domain	Specialisation
117	1,031	LIVESTOCK (1,026 = 99.5%)	Master of Flocks
104	100	PERSON (100 = 100%)	Personnel
141	83	OIL (83 = 100%)	Oil
120	84	WOOL (65 = 77%)	Wool
131	16	WHEEL (16 = 100%)	Chariot dept.
124-E	21	SPICE (21 = 100%)	Spice

Table 6. Selected scribe specialisation at Knossos.

4.4 Inca Khipu: The Cross-Medium Proof

619 khipus, 54,403 cords, 110,677 knots. Working dataset: 38,736 pendant cords with colour data, 24,545 with non-zero values. Total knot value: 8,038,367. Median cord value: 15. Max: 320,535. 56 cord colours profiled.

The MDP algorithm required **no structural modification**. The variables map directly: Commodity Logogram = Cord Colour; Numeral = Knot Count; Fractional Subunit = Knot Type; Tablet = Khipu; Findspot = Provenance.

Colour	S-knot %	L-knot %	Cords	Mean Value
W (white)	74%	20%	11,106	483
AB (light brown)	71%	22%	6,436	372
NB (strong yel. brown)	82%	15%	714	507
GG (grayish green)	77%	18%	893	513
DB (deep brown)	81%	16%	734	192
LK (BLACK)	9%	88%	529	20

Table 7. Black cord knot inversion: LK uses 88% L-type (units) vs 65-82% S-type (tens) for all others.

Hue Family	Cords	Shades	Median	Mean	Max
Brown	22,575	22	11	275	245,811
White	11,106	1	20	483	320,535
Green	1,263	5	22	427	43,372

Hue Family	Cords	Shades	Median	Mean	Max
Black	657	3	17	22	1,271
Blue	638	6	20	237	11,598
Red	132	3	36	219	6,136
Orange	67	3	9	18	121

Table 8. Cord colour value separation by Munsell hue family.

Provenance specialisation: Incahuasi (3,703 cords, mean=693), Pachacamac (4,612, mean=403), La Molina (83 cords, mean=14,920), Valle de Pisco (59 cords, 100% black), Atarco (164, mean=3). **Twist direction**: S-twist (28,500 cords, mean=376) > Z-twist (2,632, mean=222), supporting Urton's (2003) hypothesis. **Colour co-occurrence**: AB+W on 245 khipus (base palette), GG with MB on 112 (specialist category).

5. Discussion

5.1 The Administrative Segregation Principle

Across seven scripts, four continents, three media, and 4,600 years, MDP reveals a strong recurring structural pattern that we term the **Administrative Segregation Principle (ASP)**:

> *When human societies build administrative systems at scale, they tend to mathematically segregate their counting domains by commodity type. Different commodities are counted using different numeral systems, measured using different fractional subunits, recorded by different scribes, stored in different rooms, and — in the case of khipu — tied with different knot types on different-coloured cords.*

We propose ASP not as an iron law but as a convergent solution: when societies reach administrative complexity, the same mathematical architecture tends to emerge independently. The Inca Empire had zero contact with Bronze Age Crete, yet both segregated counting domains by commodity type using physically distinct markers.

5.2 Functional Ontology vs. Linguistic Translation

The Linear B control reveals two types of “meaning” in administrative texts. **Semantic meaning**: *do-e-ro* = “slave.” **Functional meaning**: *do-e-ro* = “grain-department labour asset.” Traditional decipherment seeks semantic meaning. MDP provides functional meaning. Semantic meaning tells us what the ancients were *saying*. Functional meaning tells us what they were *doing*.

5.3 The Aegean Trilogy: Script Evolution

Cretan Hieroglyphic (~2100 BCE): Seal stamps, 434 signs, KU-RO already present, livestock-dominated. **Linear A** (~1800 BCE): Full accounting — 763 words, two-tier summation, fraction divergence, room specialisation. **Linear B** (~1400 BCE): Same system in Greek — 5,823 words, 65 ideograms, measure subunits fully decoded. This trajectory mirrors Proto-Cuneiform to Sumerian cuneiform: administrative writing evolves to manage economies, not to express language.

6. Limitations and Future Work

1. **Statistical significance.** Future work: permutation tests, effect sizes, confusion matrices. Acknowledged as priority.
2. **Cretan Hieroglyphic corpus size.** Only 17/317 inscriptions have commodities.
3. **Khipu summation.** 4.6% match rate; needs top-cord analysis via PENDANT_FROM relationships.
4. **Cypro-Minoan.** Natural next target (~300 objects, no digital corpus yet).
5. **Validation expansion.** Formal administrative-context accuracy metric needed for Linear B.
6. **Community engagement.** Open-source code designed for collaboration.

7. Conclusion

MDP has now been applied to seven scripts spanning 4,600 years, four continents, and three media. The Administrative Segregation Principle appears independently across unrelated civilisations. When applied to deciphered scripts, MDP provides a functional economic ontology complementing linguistic translation. When applied to undeciphered scripts, it provides the first systematic classification of unknown signs into commodity-specific domains.

The discovery that Inca khipu cord colours function analogously to Minoan commodity ideograms suggests the mathematics of human bureaucracy is not culturally transmitted but convergently evolved. When societies reach administrative complexity, the same architecture tends to emerge, whether administrators are pressing clay in Bronze Age Crete or tying knots in the Andes.

Data Availability

All sources open-access: lineara.xyz, linearb.xyz, linear0.xyz (Hogan). OKR: DOI 10.5281/zenodo.18025748. Code: github.com/souldriver007/mdp-ancient-scripts. Paper I: DOI 10.17613/mmjdt-ba806.

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